MONASH UNIVERSITY • MCS16

GNN-Based Multi-Omics Integration for Predicting Cancer Drug Response

MCS16

SUPERVISOR Dr. Ong Huey Fang

Aditya Ansh (33520070) • Charmaine Chee (33520399)
Wasif Hussain (33621837) • Shaahid Marikar (34086242)

WASIF HUSSAIN

Agenda

01 Problem & Opportunity

Aditya Ansh



02 ---

Wasif Hussain



03 ---

Charmaine Chee



04 MVP Demo & What's Next

Shaahid Marikar



aditya

Problem Statement

01

41 Days-11 Months

Genomic testing delays faced by advanced cancer patients awaiting matched therapy decisions

02

40%+ Patients

Advanced cancer patients missing genomic testing entirely due to fragmented data pipelines

03

“Data Tsunami”

Oncologists struggle to interpret multi-omics variants without integrated bioinformatic support

Why This Project Exists

fraction of physicians who accurately interpret complex genomic variants

14.6%

The bottleneck

31%

response rate with biomarker-matched therapies vs. 10.5% standard care.

higher response when bioinformatic support is provided

Opportunity

~3x

The Gap

10.5%

response rate without matched therapy - nearly 3x lower outcomes

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Who is this for?



Cancer pharmacology researchers

Scan drug candidates against cancer cell line profiles before costly wet lab assays



Bioinformaticians

Benchmark multi omics integration and GNN approaches for drug response prediction



Academic users & students

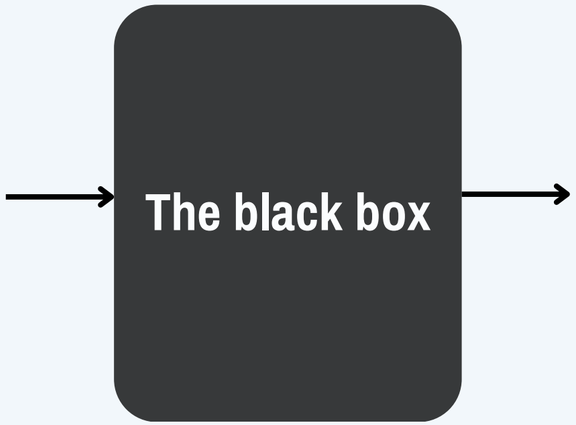
Explore how genomic features relate to drug sensitivity through our application’s dashboard

What does the system deliver

Not every cancer drug works on every cancer, and finding out the hard way costs time.

A standalone app can quickly check how likely a cancer drug is to work against a specific cancer type, before committing to further testing.

What the user puts in
The cancer: Choose a cancer profile from 500+ lab-studied cancer samples
The drug: Choose from about 500 known cancer drugs



What the user gets back
A prediction: Whether the cancer is likely to be sensitive or resistant to that drug
A confidence score: How sure the app is about that prediction
Visual summaries: Graphs that put the result in context

WASIF HUSSAIN

The black box - How it works, and how we know it works

1. What powers it

Cancer's genetic fingerprint
Several layers of genetic data

Drug's chemical makeup
How the molecule is built

AI model
Learns from thousands of lab tests

2. What's new

Same public lab data as other researchers, with new additions

Drug as a network of atoms

Several layers of genetic data combined in a novel way

3. How we proved it

Head-to-head test
Against a simpler standard model, same data

Remove one piece at a time
Shows each addition earns its place

Progress Tracking - RTM

ADITYA ANSH

Req	Description	Categories	Type	Source
R02	Delete user uploaded files after a prediction session ends, ensuring no sensitive research data is retained.	Security, Privacy	NFR	Developer
R04	Display a molecular profile summary of the top 5 mutations	UI, Visualisation	FR	Oncologist

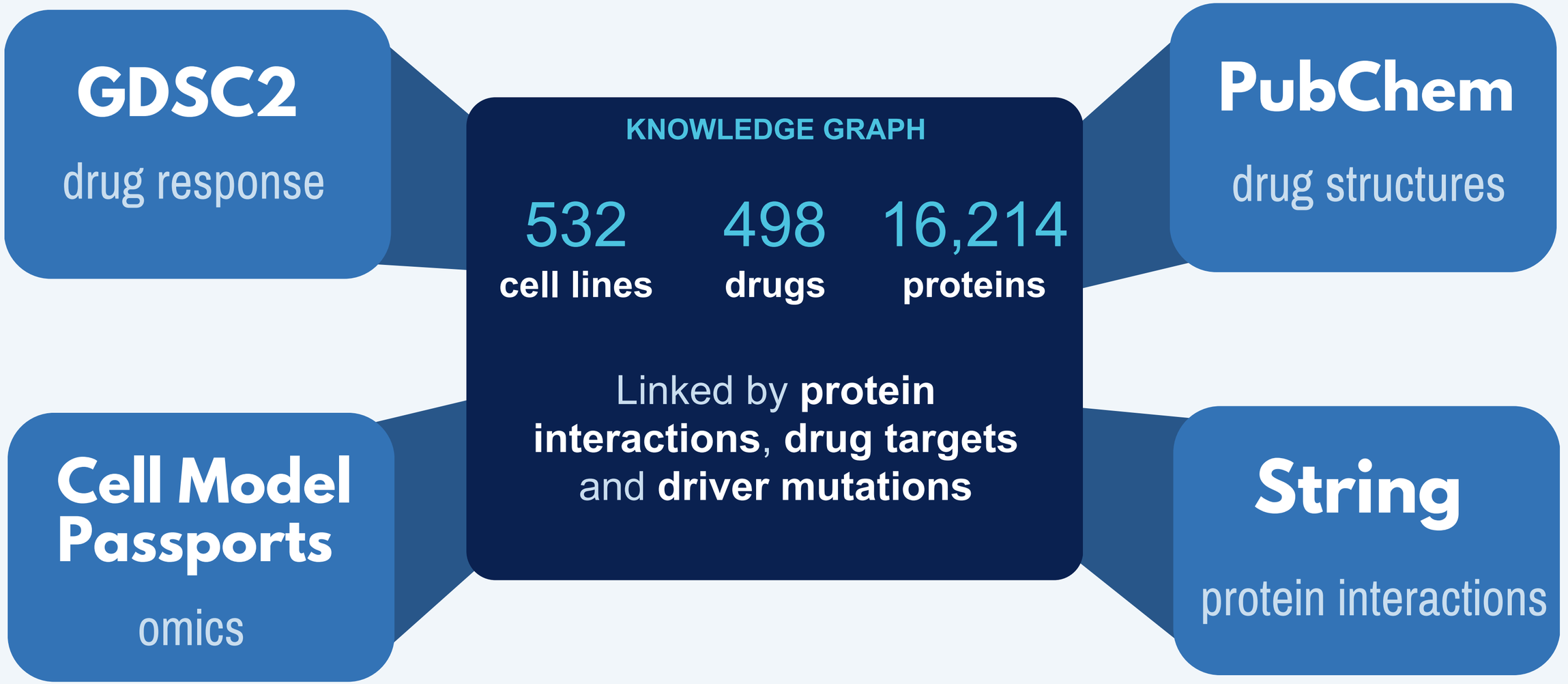
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requirements complete

1 / 3

major milestones achieved

From Public Data to a Biological Graph



Our Model

MoGraphDRP
published base, reproduced

+

Proteomics
third omics input

+

Pair attention
our novel component

The novelty: each drug's **target proteins** attend to the cell line's **mutated proteins** across the protein network

One switch per addition • tested on unseen cell lines

The Application



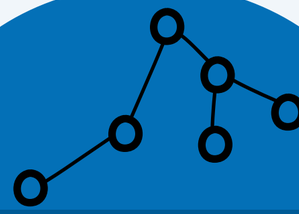
Desktop Application

PySide6



API

FastAPI



Model

PyTorch

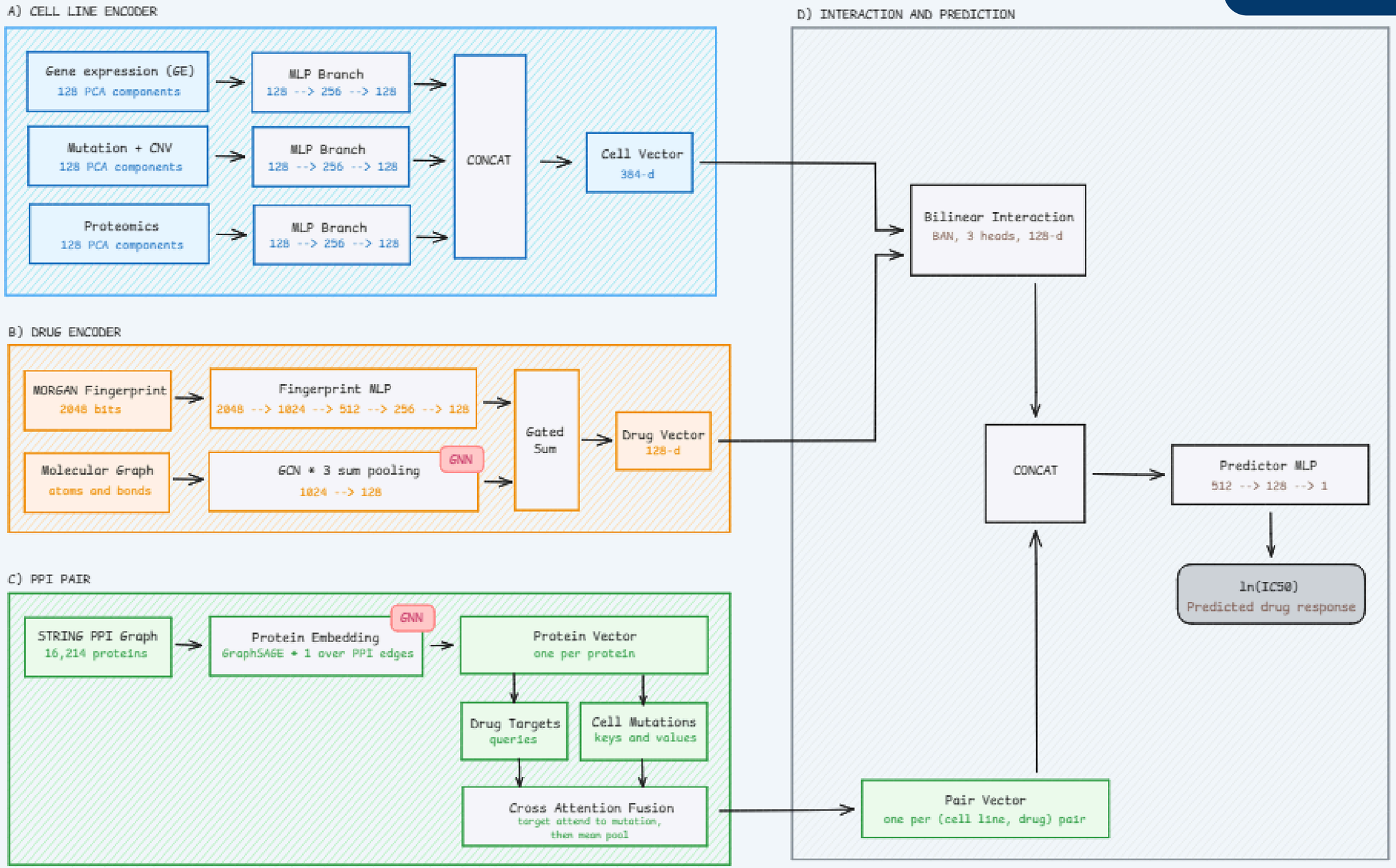
PyG

RDKit

Runs locally • trained on a laptop GPU (GTX 1650, 4 GB)

PROPOSED FINAL ARCHITECTURE

CHARMAINE CHEE



LIVE DEMO

MVP DEMO

Desktop dashboard walkthrough

01 Dataset Upload

02 Model Execution Log

03 Final Results

Real prediction from the best HeteroIC50GNN checkpoint. One function highlighted; full pipeline narration deferred.

SHAAHID MARIKAR



SHAAHID MARIKAR

Next Steps & Priorities

Improve Model Performance & Usability

01 **Implement Atomic Level Graph**

Add layer depth to improve prediction

02 **Improve User Understanding**

Add tooltip, model visualisations & notifications

CONTINUE

GNN + protein–protein interaction track

Sprint priorities and KIV items remain visible on the team Notion board.

Thank You!

References

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